REPUBLIC POLYTECHNIC

Lesson 4
Queuing Theory – Solution

E211 – Operations Planning II

Problem Statement

Figure 1: IKEA self-checkout area

Data gathered:

- Arrival rate = 150 customers per hour
- Mean checkout time = 2.5 minutes

Queuing Model

Appropriate queuing model for IKEA self-checkout is M/M/k. If number of self-checkout machines is 8, then the queuing model would be **M/M/8**.

Assumptions:

- Input source: Infinite population; no balks, no reneging
- Arrival process: Number of arrivals follows **Poisson distribution**
- Queue: Unlimited; single line
- Queue discipline: FIFO (FCFS)
- Service mechanism: **Multiple servers,** single phase system
- Service time: **Negative exponential distribution**
- Arrivals and service times are independent of the number of customers in system
- Arrival rate must be smaller than sum of service rate of servers, i.e.

  $\lambda < K\mu$ **(for K servers)**

Steady state equilibrium condition

Minimum Number of Servers, k

Arrival rate, λ = 150/60 = 2.5 per minute
Service rate, μ = 1/ 2.5 = 0.4 per minute

Steady state equilibrium holds when $\lambda < k\mu$ (for k servers)

Therefore, minimum k > λ/μ > 2.5/ 0.4 > 6.25

Minimum number of self-checkout machines, k = 7

Performance Measures

Arrival rate = 150 customers per hour
Mean checkout time = 2.5 minutes

Arrival rate	λ	2.500	per minute
Service rate	μ	0.4000	per minute
No. of servers	k	8	
Utilisation	ρ	0.7813	
Prob (no customers)	P_0	1.58E-03	
No. of customers in queue	Lq	1.4936	
No. of customers in system	Ls	7.7436	
Waiting time in queue	Wq	0.5975	minutes
Time in system	Ws	3.0975	minutes

- Utilization, ρ = 2.5/ (8 x 2.5) = 0.7813
- Average number of customers in queue, Lq = 1.4936
- Average number in system, Ls = Lq + λ/μ = 1.4936 + (2.5/ 0.4) = 7.7436
- Average waiting time Wq, = Lq/ λ = 1.4936/ 2.5 = 0.5975 min
- Average time in system Ws = Wq + $1/\mu$ = 0.5975 + (1/ 0.4) = 3.0975 mins

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Probability of at least 1 self-checkout available

No. of customers in system, n	$\frac{(\lambda/\mu)^n}{n!}$	Probability of n customers in system, Pn	
		$\frac{(\lambda/\mu)^n}{n!}P_0$ for $0 \le n \le k$	$\frac{(\lambda/\mu)^n}{k!k^{n-k}}P_0$ for $n > k$
0	1	1.584E-03	0.659221436
1	6.25	9.902E-03	0.515016747
2	19.53125	3.094E-02	0.402356834
3	40.690104	6.446E-02	0.314341276
4	63.578288	1.007E-01	0.245579122
5	79.47286	1.259E-01	0.191858689
6	82.784229	1.312E-01	0.149889601
7	73.91449	1.171E-01	0.117101251
8	57.745695	9.149E-02	0.091485352
9	40.101177	6.353E-02	0.071472931
10	25.063236	3.971E-02	0.055838228
11	14.240475	2.256E-02	0.043623615
12	7.416914	1.175E-02	0.034080949

Sum these values for $\sum_0^7 \frac{(\frac{\lambda}{\mu})^n}{n!} P_0$

No. of self-checkout machines, k = 8

Probability of at least 1 self-checkout machine is available,

= Probability of 7 or less customers in system

= P(n<=7) = $\sum_0^7 \frac{(\frac{\lambda}{\mu})^n}{n!} P_0$ = 0.582

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Probability of at least 2 customers in queue

No. of customers in system, n	$\frac{(\lambda/\mu)^n}{n!}$	Probability of n customers in system, Pn	
		$\frac{(\lambda/\mu)^n}{n!}P_0$ for $0 \le n \le k$	$\frac{(\lambda/\mu)^n}{k!k^{n-k}}P_0$ for $n > k$
0	1	1.584E-03	0.659221436
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No. of self-checkout machines, k = 8

Sum these values for $\sum_0^9 \frac{(\frac{\lambda}{\mu})^n}{n!} P_0$

Probability of at least 2 customers waiting in the queue,

= Probability of 10 or more customers in system

= P(n>=10) = 1- $\sum_0^9 \frac{(\frac{\lambda}{\mu})^n}{n!} P_0$ = 0.255

Determining Service Time

Suppose that there is a cashier counter set up to assist customers who are unable to perform self-checkout. Your task is to determine an average service time standard for the cashier checkout so that the service level and staff utilization are reasonable. Initial estimates are that on average, 20 customers are expected to use the cashier checkout every hour.

Discuss the approach you would take to carry out your task.

Determining Service Time – M/G/1 model

Make use of M/G/1 model. Experiment with different values of Service Time to find out the corresponding service level.

M/G/1

Arrival Rate, λ	0.333	per minute
Service Rate, μ	0.400	per minute
Service Time, 1/μ	2.5	minute
Standard deviation of service times, σ	0.500	minute
System Utilization, ρ (set target)	0.833	
Probability of no customers, P_0	0.167	
Average number in queue, Lq	2.167	
Average time in queue, Wq	6.500	minutes
Average number in system, Ls	3.000	
Average time in system, Ws	9.000	minutes

Average service time can be regulated through adequate staff training and limiting the number of check out items for each customer.

Average service time impacts utilization and service level

Standard deviation may be adjusted with more data

Is utilization of 0.833 reasonable?

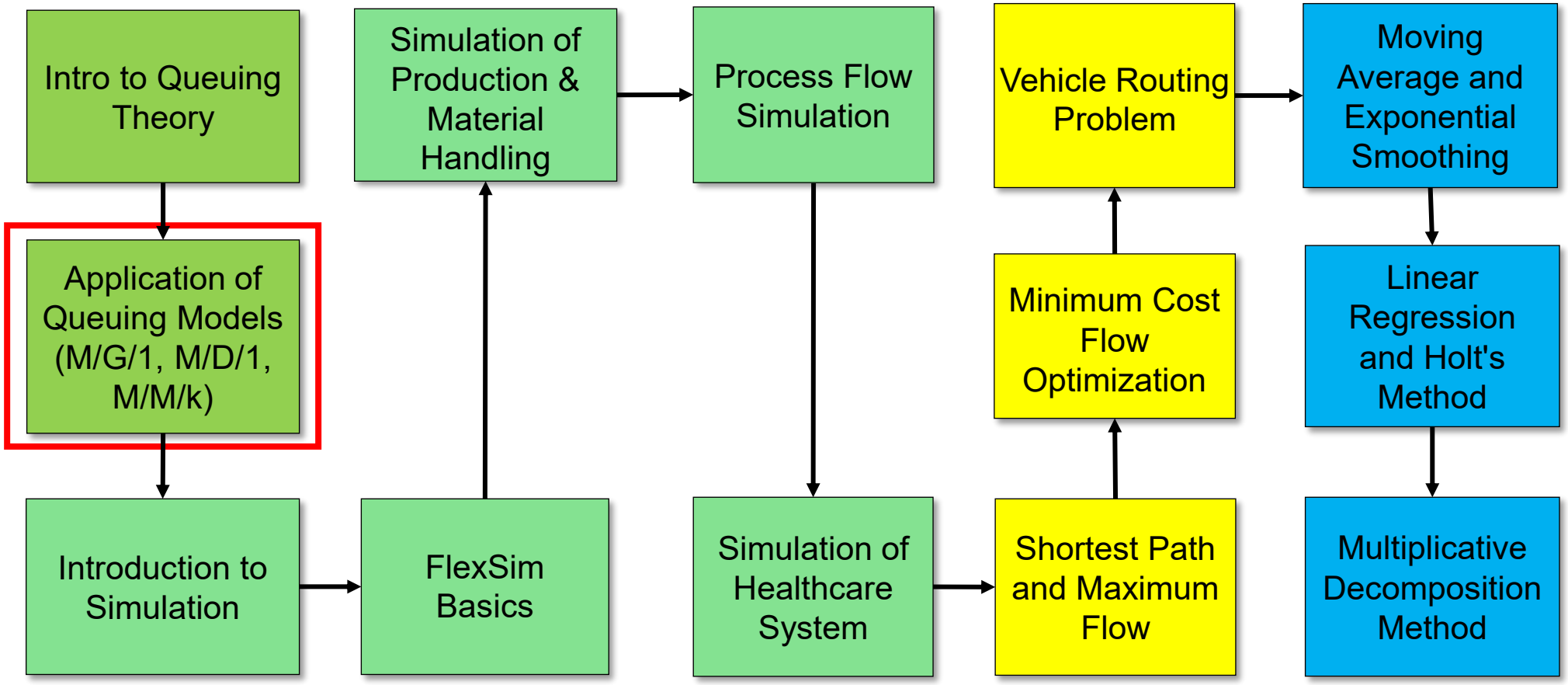
Are the average waiting time and time in system ok? Experiment with the service time to find a desirable service level.

Learning Outcomes

At the end of the lesson, students should be able to:

- Calculate performance measures of M/M/k queuing models.
- Build Excel models of queuing systems.
- Apply queuing models in the design of service operations.

E211 Progression



Appendix – Calculation of Lq

Average number in queue Lq = $\frac{\left(\frac{\lambda}{\mu}\right)^{k}\lambda\mu}{(k-1)!(k\mu-\lambda)\text{^}2}$ *P_0

$= \frac{(\frac{2.5}{0.4})^{8}*2.5*0.4}{(8-1)!(8*0.4-2.5)\text{^}2} * 0.158$

= 1.4936

P_0 is determined in the M/M/k template in Excel.